Notice that all the comparison operators in the set $\{=, <, \leq, >, \geq, \neq\}$ can apply to attributes whose domains are *ordered values*, such as numeric or date domains. Domains of strings of characters are also considered to be ordered based on the collating sequence of the characters. If the domain of an attribute is a set of *unordered values*, then only the comparison operators in the set $\{=, \neq\}$ can be used. An example of an unordered domain is the domain Color = { 'red', 'blue', 'green', 'white', 'yellow', ...}, where no order is specified among the various colors. Some domains allow additional types of comparison operators; for example, a domain of character strings may allow the comparison operator SUBSTRING_OF.

Division

Let A have 2 fields, x and y; B have only field y:

- $A/B = \{\langle x \rangle \mid \exists \langle x, y \rangle \in A \ \forall \langle y \rangle \in B\}$

i.e., A/B contains all x tuples such that for every y tuple in B, there is an xy tuple in A.

Or: If the set of y values associated with an x value in A contains all y values in B, the x value is in A/B.

Query contain **every** or **all**

The division operator is useful for expressing certain kinds of queries, for example: Find the names of sailors who have reserved **all** boats.

We discuss division through an example. Consider two relation instances A and B in which A has (exactly) two fields x and y and B has just one field y, with the same domain as in A. We define the *division* operation A/B as the set of all x values (in the form of unary tuples) such that for *every* y value in (a tuple of) B, there is a tuple $\langle x,y \rangle$ in A.

Another way to understand division is as follows. For each x value in (the first column of) A, consider the set of y values that appear in (the second field of) tuples of A with that x value. If this set contains (all y values in) B, the x value is in the result of A/B.

Division is illustrated in Figure 4.14. It helps to think of A as a relation listing the parts supplied by suppliers, and of the B relations as listing parts. A/Bi computes suppliers who supply *all* parts listed in relation instance Bi.

A

sno	*pno*
s1	p1
s1	p2
s1	p3
s1	p4
s2	p1
s2	p2
s3	p2
s4	p2
s4	p4

B1

pno
p2

B2

pno
p2
p4

B3

pno
p1
p2
p4

A/B1

sno
s1
s2
s3
s4

A/B2

sno
s1
s4

A/B3

sno
s1

Figure 4.14 Examples Illustrating Division

Expressing A/B in terms of the basic algebra operators:

Idea: For A/B, compute all x values that are not \`disqualified' by some y value in B.

- x value is *disqualified* if by attaching y value from B, we obtain an xy tuple that is not in A.

Disqualified x values: $\pi_x((\pi_x(A) \times B) - A)$

A/B: $\pi_x(A)$ − all disqualified tuples

The basic idea is to compute all x values in A that are not disqualified. An x value is disqualified if by attaching a y value from B, we obtain a tuple <x,y> that is not in A. We can compute disqualified tuples using the algebra expression

$$\pi_x((\pi_x(A) \times B) - A)$$

Thus we can define A/B as

$$\pi_x(A) - \pi_x((\pi_x(A) \times B) - A)$$

Q. Find Sid's who enrolled for **every** course

Stu

Sid	Cid
s_1	c_1
s_1	c_2
s_2	c_1
s_2	c_2
s_3	c_3
s_1	c_3

Course

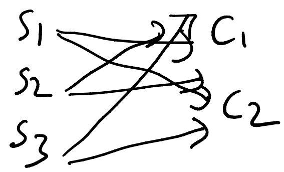
cid
c_1
c_2

$\overset{A}{Stu}(\overset{x}{Sid}, \overset{y}{cid}) / \overset{B}{Course}(\overset{y}{cid})$

Sid
s_1
s_2

$A(x,y)/B(y) = \pi_x(A) - \pi_x((\pi_x(A) \times B) - A)$

$\pi_x(A)$ Course



$$\boxed{\pi_x(\pi_x(A)\times B) - A)}$$

$\pi_x(A) \times B =$

s_1	c_1
s_1	c_2
s_2	c_1
s_2	c_2
s_3	c_1
s_3	c_2

$-$

A

s_1	c_1
s_1	c_2
s_2	c_1
s_2	c_2
s_3	c_3
s_1	c_3

$=$

π_{sid}

s_3	c_1
s_3	c_2

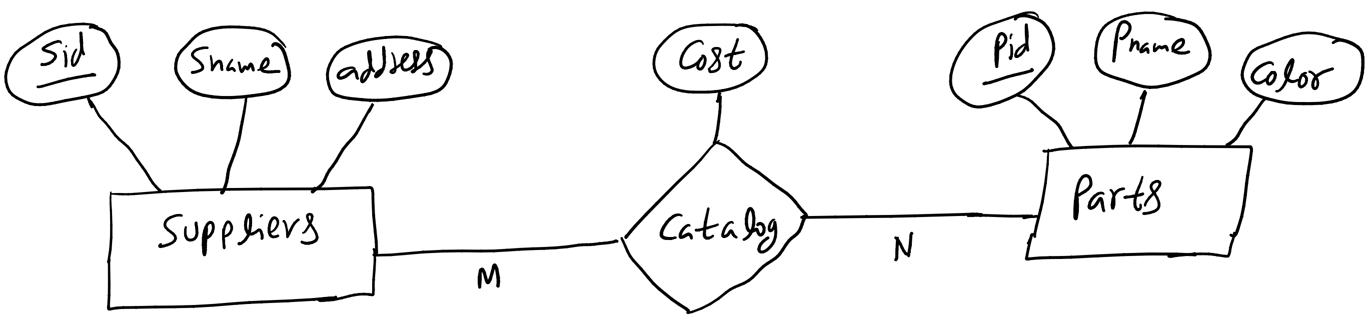
$= \boxed{s_3}$

$$\begin{array}{c} s_1 \\ s_2 \\ s_3 \end{array} - s_3 = \boxed{\begin{array}{c} s_1 \\ s_2 \end{array}}$$

Exercise Consider the following schema:

Suppliers(*sid:* integer, *sname:* string, *address:* string)
Parts(*pid:* integer, *pname:* string, *color:* string)
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

The key fields are underlined, and the domain of each field is listed after the field name. Thus *sid* is the key for Suppliers, *pid* is the key for Parts, and *sid* and *pid* together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra:



Suppliers(*sid:* integer, *sname:* string, *address:* string)

Parts(*pid:* integer, *pname:* string, *color:* string)

F.K F.K
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

Find the names of suppliers who supply some red part.

$$\pi_{sname}(\pi_{sid}((\pi_{pid}\sigma_{color='red'}Parts) \bowtie Catalog) \bowtie Suppliers)$$

$$\pi_{name}(\sigma_{Color='Red'}(Parts \bowtie Catalog \bowtie Suppliers))$$

Suppliers(*sid:* integer, *sname:* string, *address:* string)
Parts(*pid:* integer, *pname:* string, *color:* string)
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

Find the sids of suppliers who supply some red or green part.

$$\pi_{sid}(\pi_{pid}(\sigma_{color='red' \vee color='green'} Parts) \bowtie catalog)$$

$$\pi_{sid}\left(\sigma_{(Col=Red \vee Col=green)}(Parts \bowtie Catalog)\right)$$

$$\rho(T_1,\ \pi_{sid}(\pi_{pid}(\sigma_{Col=Red}(Parts)) \bowtie Catalog))$$

$$\rho(T_2,\ \pi_{sid}(\pi_{pid}(\sigma_{Col=green}(Parts)) \bowtie Catalog)).$$

$$(T_1 \cup T_2)$$

Suppliers(*sid:* `integer`, *sname:* `string`, *address:* `string`)
Parts(*pid:* `integer`, *pname:* `string`, *color:* `string`)
Catalog(*sid:* `integer`, *pid:* `integer`, *cost:* `real`)

Find the sids of suppliers who supply some red part and some green part.

$$\boxed{\pi_{sid}\left(\sigma_{(Col=Red \cap Col=green)}(Parts \bowtie Catalog)\right)}$$

X

This solution is incorrect. A part can be only one color (part is not a multivalued attribute); this query will always return an empty relation. The color of part can be green or red, but not both.

$\rho(T_1, \pi_{sid}(\pi_{pid}(\sigma_{Col=Red}(Parts)) \bowtie Catalog))$

$\rho(T_2, \pi_{sid}(\pi_{pid}(\sigma_{Col=green}(Parts)) \bowtie Catalog))$.

$(T_1 \cap T_2)$

✓

$\rho(R1, \pi_{sid}((\pi_{pid}\sigma_{color='red'}Parts) \bowtie Catalog))$
$\rho(R2, \pi_{sid}((\pi_{pid}\sigma_{color='green'}Parts) \bowtie Catalog))$
$R1 \cap R2$

✓

Suppliers(<u>*sid:* integer</u>, *sname:* string, *address:* string)
Parts(<u>*pid:* integer</u>, *pname:* string, *color:* string)
Catalog(<u>*sid:* integer, *pid:* integer</u>, *cost:* real)

Find the sids of suppliers who supply some red part or are at 221 Packer Ave.

$\rho(R1, \pi_{sid}((\pi_{pid}\sigma_{color='red'}Parts) \bowtie Catalog))$
$\rho(R2, \pi_{sid}\sigma_{address='221PackerStreet'}Suppliers)$
$R1 \cup R2$

✓

Suppliers(<u>*sid:* integer</u>, *sname:* string, *address:* string)
Parts(<u>*pid:* integer</u>, *pname:* string, *color:* string)
Catalog(<u>*sid:* integer, *pid:* integer</u>, *cost:* real)

Find the sids of suppliers who supply <u>every</u> part.

$\boxed{(\pi_{sid,pid}Catalog)/(\pi_{pid}Parts)}$

Find the sids of suppliers who supply every red part.

$\boxed{(\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='red'}Parts)}$

Find the sids of suppliers who supply every red or green part.
Find the sids of suppliers who supply every red part or supply every green part.

not same
why?

$(\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='red' \vee color='green'}Parts)$

not same

why ?

$\rho(R1, ((\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='red'}Parts)))$
$\rho(R2, ((\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='green'}Parts)))$
$R1 \cup R2$

https://www.cs.colostate.edu/~cs430dl/yr2016su/more_examples/Ch6/Relational%20algebra%20queries.pdf

Find the sids of suppliers who supply every red or green part.

$$(\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='red' \vee color='green'}Parts)$$

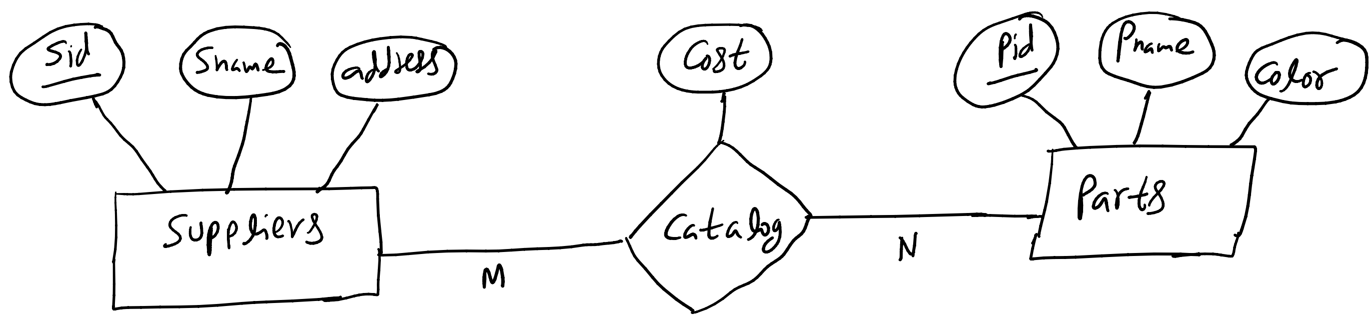
Find the sids of suppliers who supply every red part or supply every green part.

$$\rho(R1, ((\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='red'}Parts)))$$
$$\rho(R2, ((\pi_{sid,pid}Catalog)/(\pi_{pid}\sigma_{color='green'}Parts)))$$
$$R1 \cup R2$$

Exercise Consider the following schema:

Suppliers(*sid:* integer, *sname:* string, *address:* string)
Parts(*pid:* integer, *pname:* string, *color:* string)
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

The key fields are underlined, and the domain of each field is listed after the field name. Thus *sid* is the key for Suppliers, *pid* is the key for Parts, and *sid* and *pid* together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra:



Suppliers(*sid:* integer, *sname:* string, *address:* string)

Parts(*pid:* integer, *pname:* string, *color:* string)

F.K F.K

Catalog(*sid:* integer, *pid:* integer, *cost:* real)

Suppliers(*sid:* integer, *sname:* string, *address:* string)
Parts(*pid:* integer, *pname:* string, *color:* string)
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

Find pairs of sids such that the supplier with the first sid charges more for some part than the supplier with the second sid.

$\rho(R1, Catalog)$

$\rho(R2, Catalog)$

$\pi_{R1.sid,R2.sid}(\sigma_{R1.pid=R2.pid \wedge R1.sid \neq R2.sid \wedge R1.cost>R2.cost}(R1 \times R2))$

Suppliers($\underline{sid: \texttt{integer}}$, $sname$: string, $address$: string)
Parts($\underline{pid: \texttt{integer}}$, $pname$: string, $color$: string)
Catalog($\underline{sid: \texttt{integer}, pid: \texttt{integer}}$, $cost$: real)

Find the pids of parts that are supplied by at least two different suppliers.

$$\rho(R1, Catalog)$$
$$\rho(R2, Catalog)$$
$$\pi_{R1.pid}\sigma_{R1.pid=R2.pid \wedge R1.sid \neq R2.sid}(R1 \times R2)$$

Suppliers($\underline{sid: \texttt{integer}}$, $sname$: string, $address$: string)
Parts($\underline{pid: \texttt{integer}}$, $pname$: string, $color$: string)
Catalog($\underline{sid: \texttt{integer}, pid: \texttt{integer}}$, $cost$: real)

Find the pids of the most expensive parts supplied by suppliers named Yosemite Sham.

$$\rho(R1, \pi_{sid}\sigma_{sname='YosemiteSham'}Suppliers)$$
$$\rho(R2, R1 \bowtie Catalog)$$
$$\rho(R3, R2)$$
$$\rho(R4(1 \rightarrow sid, 2 \rightarrow pid, 3 \rightarrow cost), \sigma_{R3.cost<R2.cost}(R3 \times R2))$$
$$\pi_{pid}(R2 - \pi_{sid,pid,cost}R4)$$

In-class Exercise Relational Algebra
Consider the following schema:

Suppliers(<u>*sid:* integer</u>, *sname:* string, *address:* string)
Parts(<u>*pid:* integer</u>, *pname:* string, *color:* string)
Catalog(<u>*sid:* integer, *pid:* integer</u>, *cost:* real)

The key fields are underlined, and the domain of each field is listed after the field name. Therefore *sid* is the key for Suppliers, *pid* is the key for Parts, and *sid* and *pid* together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra.

1. Find the *name*s of suppliers who supply some red part.

2. Find the *sid*s of suppliers who supply some red or green part.

3. Find the *sid*s of suppliers who supply some red part or are at 221 Packer Street.

4. Find the *sid*s of suppliers who supply some red part and some green part.

5. Find the *sid*s of suppliers who supply every part.

6. Find the *sid*s of suppliers who supply every red part.

7. Find the *sid*s of suppliers who supply every red or green part.

8. Find the *sid*s of suppliers who supply every red part or supply every green part.

9. Find pairs of *sid*s such that the supplier with the first *sid* charges more for some part than the supplier with the second *sid*.

10. Find the *pid*s of parts supplied by at least two different suppliers.

11. Find the *pid*s of the most expensive parts supplied by suppliers named Yosemite Sham.

Consider the following schema:

Suppliers(*sid:* integer, *sname:* string, *address:* string)
Parts(*pid:* integer, *pname:* string, *color:* string)
Catalog(*sid:* integer, *pid:* integer, *cost:* real)

The key fields are underlined, and the domain of each field is listed after the field name. Therefore *sid* is the key for suppliers, *pid* is the key for Parts, and *sid* and *pid* together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra.

1. Find the *names* of suppliers who supply some red part.

$\pi_{sname}(\pi_{sid}((\pi_{pid}\sigma_{color=\_'red'\_}Parts) \bowtie Catalog) \bowtie Suppliers)$

Catalog

SID	PID	Cost
1	1	$10.00
1	2	$20.00
1	3	$30.00
1	4	$40.00
1	5	$50.00
2	1	$9.00
2	3	$34.00
2	5	$48.00

Parts

PID	Pname	Color
1	Red1	Red
2	Red2	Red
3	Green1	Green
4	Blue1	Blue
5	Red3	Red

SID	Sname	Address
1	Yosemite Sham	Devil's canyon, AZ
2	Wiley E. Coyote	RR Asylum, NV
3	Elmer Fudd	Carrot Patch, MN

Ok, let's break this down.

$\sigma_{color=\_'red'\_}$*Parts gives us*

PID	Pname	Color
1	Red1	Red
2	Red2	Red
5	Red3	Red

$\pi_{pid}\sigma_{color=\_'red'\_}Parts$ gives us

PID
1
2
5

$((\pi_{pid}\sigma_{color=\_'red'\_}Parts) \bowtie Catalog)$ gives us:

SID	PID	Cost
1	1	$10.00
1	2	$20.00
1	5	$50.00
2	1	$9.00
2	5	$48.00

$\pi_{sid}((\pi_{pid}\sigma_{color=\_'red'\_}Parts) \bowtie Catalog)$ gives us:

SID
1
2

$\pi_{sid}((\pi_{pid}\sigma_{color=\_'red'\_}Parts) \bowtie Catalog) \bowtie Suppliers)$ gives us:

SID	Sname	Address
1	Yosemite Sham	Devil's canyon, AZ
2	Wiley E. Coyote	RR Asylum, NV

And finally $\pi_{sname}(\pi_{sid}((\pi_{pid}\sigma_{color=\_'red'\_}Parts) \bowtie Catalog) \bowtie Suppliers)$

Gives us:

Sname
Yosemite Sham
Wiley E. Coyote

2. Find the *sid*s of suppliers who supply some red or green part.

$\pi_{sid}(\pi_{pid}(\sigma_{color='red' \vee color='green'} Parts) \bowtie catalog)$

3. Find the *sid*s of suppliers who supply some red part or are at 221 Packer Street.

$\rho(R1, \pi_{sid}((\pi_{pid}\sigma_{color='red'}Parts) \bowtie Catalog))$
$\rho(R2, \pi_{sid}\sigma_{address=''221PackerStreet'}Suppliers)$
$R1 \cup R2$

4. Find the *sid*s of suppliers who supply some red part and some green part.

$\rho(R1, \pi_{sid}((\pi_{pid}\ \sigma_{color='red'} Parts) \bowtie| Catalog))$
$\rho(R2, \pi_{sid}((\pi_{pid}\ \sigma_{color='green'} Parts) \bowtie Catalog))$
$R1 \cap R2$

5. Find the *sid*s of suppliers who supply every part.

$(\pi_{sid,pid}\, Catalog)/(\pi_{pid}\, Parts)$

Given:

Parts

PID	Pname	Color
1	Red1	Red
2	Red2	Red
3	Green1	Green
4	Blue1	Blue
5	Red3	Red

Catalog

SID	PID	Cost
1	1	$10.00
1	2	$20.00
1	3	$30.00
1	4	$40.00
1	5	$50.00
2	1	$9.00
2	3	$34.00
2	5	$48.00
3	1	$11.00

π_{pid} *Parts gives us:*

PID
1
2
3
4
5

$\pi_{sid,pid}$ *Catalog gives us:*

SID	PID
1	1
1	2
1	3
1	4
1	5
2	1
2	3
2	5
3	1

$(\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ Parts)$

Asks the question – what sids in catalog contain all the part numbers in the divisor. There is only one sid that has all the part numbers, 1.

6. Find the *sid*s of suppliers who supply every red part.

$(\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ \sigma_{color='red'}\ Parts)$

7. Find the *sid*s of suppliers who supply every red or green part.

$(\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ \sigma_{color='red' \vee color='green'}\ Parts)$

Let's look at this one:

Catalog

SID	PID	Cost
1	1	$10.00
1	2	$20.00
1	3	$30.00
1	4	$40.00
1	5	$50.00
2	1	$9.00
2	3	$34.00
2	5	$48.00
3	1	$11.00

Parts

PID	Pname	Color
1	Red1	Red
2	Red2	Red
3	Green1	Green
4	Blue1	Blue
5	Red3	Red

$\sigma_{color='red' \vee color='green'}$ *Parts gives us*
1,2,3,5

$(\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ \sigma_{color='red' \vee color='green'}\ Parts)$ then says :

What supplier supplies all 4 of those parts?
The answer is only supplier 1.

8. Find the *sid*s of suppliers who supply every red part or supply every green part.

$\rho(R1,\ ((\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ \sigma_{color='red'}\ Parts)))$
$\rho(R2,\ ((\pi_{sid,pid}\ Catalog)/(\pi_{pid}\ \sigma_{color='green'}\ Parts)))$
$R1 \cup R2$

OK – lets look at this one then:

Catalog

SID	PID	Cost
1	1	$10.00
1	2	$20.00
1	3	$30.00
1	4	$40.00
1	5	$50.00
2	1	$9.00
2	3	$34.00
2	5	$48.00
3	1	$11.00

Parts

PID	Pname	Color
1	Red1	Red
2	Red2	Red
3	Green1	Green
4	Blue1	Blue
5	Red3	Red

$\pi_{pid}\ \sigma_{color='red'}$ *Parts gives us 1,2,5*

$\pi_{sid,pid}$ *Catalog)/(*$\pi_{pid}\ \sigma_{color='red'}$ *Parts) gives us 1*

$\pi_{pid}\ \sigma_{color='green'}$ *Parts gives us 3*

$\pi_{sid,pid}$ *Catalog)/(*$\pi_{pid}\ \sigma_{color='green'}$ *Parts*) gives us 1,2

The union of the 2 give us 1,2

9. Find pairs of *sid*s such that the supplier with the first *sid* charges more for some part than the supplier with the second *sid*.

$\rho(R1, Catalog)$
$\rho(R2, Catalog)$
$\pi_{R1.sid, R2.sid}(\sigma_{R1.pid=R2.pid \wedge R1.sid \mathrel{!}= R2.sid \wedge R1.cost > R2.cost}(R1 \times R2))$

Let's look at an example of this one:

Catalog:

SID	PID	Cost
1	1	$10.00
2	1	$9.00
2	3	$34.00
3	1	$11.00

R1 x R2 gives us:

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	1	1	$10.00
1	1	$10.00	2	1	$9.00
1	1	$10.00	2	3	$34.00
1	1	$10.00	3	1	$11.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	2	1	$9.00
2	1	$9.00	2	3	$34.00
2	1	$9.00	3	1	$11.00
2	3	$34.00	1	1	$10.00
2	3	$34.00	2	1	$9.00
2	3	$34.00	2	3	$34.00
2	3	$34.00	3	1	$11.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00
3	1	$11.00	2	3	$34.00
3	1	$11.00	3	1	$11.00

At this point, we are selecting for the 3 and clauses. The first ($\sigma_{R1.pid=R2.pid}$) gives us:

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	1	1	$10.00
1	1	$10.00	2	1	$9.00
1	1	$10.00	3	1	$11.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	2	1	$9.00
2	1	$9.00	3	1	$11.00
2	3	$34.00	2	3	$34.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00
3	1	$11.00	3	1	$11.00

The second and clause ($\sigma_{R1.pid=R2.pid \wedge R1.sid \,!=\, R2.sid}$) *gives us:*

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	3	1	$11.00
1	1	$10.00	2	1	$9.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	3	1	$11.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00

Adding in the third clause ($\sigma_{R1.pid=R2.pid \wedge R1.sid \,!= R2.sid \wedge R1.cost > R2.cost}$) gives us:

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	2	1	$9.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00

And finally, projecting the pairs gives us:

SID	SID
1	2
3	1
3	2

10. Find the *pid*s of parts supplied by at least two different suppliers.

$\rho(R1, Catalog)$
$\rho(R2, Catalog)$
$\pi_{R1.pid}\,\sigma_{R1.pid=R2.pid \wedge R1.sid \,!=\, R2.sid}\,(R1 \times R2)$

Using the following:

SID	PID	Cost
1	1	$10.00
2	1	$9.00
2	3	$34.00
3	1	$11.00

R1 x R2 gives us:

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	1	1	$10.00
1	1	$10.00	2	1	$9.00
1	1	$10.00	2	3	$34.00
1	1	$10.00	3	1	$11.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	2	1	$9.00
2	1	$9.00	2	3	$34.00
2	1	$9.00	3	1	$11.00
2	3	$34.00	1	1	$10.00
2	3	$34.00	2	1	$9.00
2	3	$34.00	2	3	$34.00
2	3	$34.00	3	1	$11.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00
3	1	$11.00	2	3	$34.00
3	1	$11.00	3	1	$11.00

$\sigma_{R1.pid=R2.pid}$ *gives us:*

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	1	1	$10.00
1	1	$10.00	2	1	$9.00
1	1	$10.00	3	1	$11.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	2	1	$9.00
2	1	$9.00	3	1	$11.00
2	3	$34.00	2	3	$34.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00
3	1	$11.00	3	1	$11.00

$\sigma_{R1.pid=R2.pid \wedge R1.sid \,!= R2.sid}$ *gives us:*

SID	PID	Cost	SID	PID	Cost
1	1	$10.00	2	1	$9.00
1	1	$10.00	3	1	$11.00
2	1	$9.00	1	1	$10.00
2	1	$9.00	3	1	$11.00
3	1	$11.00	1	1	$10.00
3	1	$11.00	2	1	$9.00

Projecting on PID gives us a single part number – 1 (eliminating the duplicates)

11. Find the *pid*s of the most expensive parts supplied by suppliers named Yosemite Sham.

$\rho(R1, \pi_{sid}\, \sigma_{sname='YosemiteSham'}\, Suppliers)$
$\rho(R2, R1 \bowtie Catalog)$
$\rho(R3, R2)$
$\rho(R4(1 \rightarrow sid, 2 \rightarrow pid, 3 \rightarrow cost), \sigma_{R3.cost < R2.cost}(R3 \times R2))$
$\pi_{pid}(R2 - \pi_{sid,pid,cost}\, R4)$

Given:

Suppliers

SID	Sname	Address
1	Wiley E. Coyote	Acme Testing Ground, NV
2	Yosemite Sham	Devil's Canyon, AZ
3	Elmer Fudd	Carrot Patch, MN

and Catalog:

SID	PID	Cost
1	1	$10.00
2	1	$9.00
2	2	$21.00
2	3	$34.00
3	1	$11.00

$\rho(R1, \pi_{sid}\, \sigma_{sname='YosemiteSham'}\, Suppliers)$

Gives us the value 2.

$\rho(R2, R1 \bowtie Catalog)$

Gives us:

SID	PID	Cost
2	1	$9.00
2	2	$21.00
2	3	$34.00

Let's look at:
$\rho(R4(1 \rightarrow sid, 2 \rightarrow pid, 3 \rightarrow cost), \sigma_{R3.cost < R2.cost}(R3 \times R2))$

R3 x R2

SID	PID	Cost	SID	PID	Cost
2	1	$9.00	2	1	$9.00
2	1	$9.00	2	2	$21.00
2	1	$9.00	2	3	$34.00
2	2	$21.00	2	1	$9.00
2	2	$21.00	2	2	$21.00
2	2	$21.00	2	3	$34.00
2	3	$34.00	2	1	$9.00
2	3	$34.00	2	2	$21.00
2	3	$34.00	2	3	$34.00

$\sigma_{R3.cost < R2.cost}(R3 \times R2)$ gives us:

SID	PID	Cost	SID	PID	Cost
2	1	$9.00	2	2	$21.00
2	1	$9.00	2	3	$34.00
2	2	$21.00	2	3	$34.00

$\rho(R4(1 \rightarrow sid, 2 \rightarrow pid, 3 \rightarrow cost), \sigma_{R3.cost < R2.cost}(R3 \times R2))$

Gives us:

SID	PID	Cost
2	1	$9.00
2	2	$21.00

$R2 - \pi_{sid,pid,cost} R4$ gives us:

SID	PID	Cost
2	3	$34.00

And projecting the PID gives us 3 as Yosemite Sham's most expensive part.